

PassportSnap

Passport Photo Web App

Product & Engineering Specification — Version 2.0 — May 2026

**The entire product in one sentence:
Upload photo → AI fixes everything → Print. That's it.**

1. Product Vision & Core Philosophy

PassportSnap is a free, browser-based passport photo tool built for one type of user: someone who is not comfortable with computers, does not know what DPI means, and just needs a correct passport photo printed right now.

1.1 The North Star Rule

A non-technical photo studio operator should be able to go from 'photo uploaded' to 'print sheet ready' in under 60 seconds, without reading any instructions.

1.2 What This App Is NOT

- NOT a photo editor — no layers, no filters, no complex tools
- NOT a technical tool — no DPI settings, no pixel dimensions, no head-ratio percentages shown to users
- NOT a subscription — free forever, no account, no cloud
- NOT dependent on the internet after first load — works fully offline

1.3 What This App IS

- A 5-step wizard any person can follow without training
- An AI tool that automatically fixes the photo — crop, align face, remove background
- A print sheet builder: multiple people on one page, drag to rearrange, set counts per person
- A download tool: print-ready PDF and digital-submission JPG in one click

1.4 The One Rule for Every Design Decision

If a non-technical user would need to ask 'what does this mean?' — remove it, rename it, or hide it.

1.5 Target Users

Who	Where	Their Goal
Photo studio operator	Tier-2/3 city India (Raipur, etc.)	Print 8–10 correct passport photos in 2 minutes
Individual applicant	At home on a laptop	Make a passport photo without visiting a studio
CSC / Govt center	Common Service Centre	Handle multiple customers' photos on one sheet, cut paper cost
Visa agent	Office with a printer	Make visa photos for different countries quickly

2. The 5-Step User Workflow

This is the entire app. Five steps. No optional paths, no advanced mode by default.

Step	Screen	What User Sees	User Action	AI Does Silently
1	Photo type	3–4 big visual cards	Tap one card	Remembers choice forever
2	Upload	Large drag-drop zone	Drop or pick photo	Detects face, auto-crops, auto-aligns
3	Check & adjust	Preview with face box and 3 sliders	Move box if wrong. Adjust brightness.	Runs face landmark detection continuously
4	Background	2–3 big image cards	Tap one card	Removes background, fills chosen color
5	Print sheet	Sheet preview, person list, drag interface	Set counts, drag if needed, Download	Tiles photos at 300 DPI into print sheet

The app remembers the last-used photo type and background choice. On second visit, the user starts directly at Step 2.

3. Screen-by-Screen Specification

3.1 Step 1 — Photo Type

Select the country/size template. Shown only on first visit or when the user explicitly changes it.

UI Element	Detail
Card grid	3–4 large cards. Each shows a photo silhouette in the correct aspect ratio, a plain-language name, and country names below. No mm, no px.
India / USA	51x51 mm. Label: 'India / USA — Most common'. Pre-selected on first load.
UK / Europe	35x45 mm. Label: 'UK, Europe, Australia'.
Canada	50x70 mm. Label: 'Canada'.
Custom	Opens a simple cm width x height input. For advanced users only.
Remember choice	Saved to localStorage. Second visit skips this step entirely.

3.2 Step 2 — Upload Photo

Accept the user's photo and run AI auto-processing in the background.

UI Element	Detail
Upload zone	Entire center of screen is the drop zone. Label: 'Drop photo here, or click to choose'. Large, obvious.
Accepted formats	JPG, PNG, HEIC, WebP. Max 20 MB. Friendly error if wrong: 'That file type doesn't work — try a JPG or PNG photo'.
After upload	Show photo as thumbnail immediately. Show progress: 'Finding your face...' then 'Done ✓'. Auto-advance to Step 3.
If no face found	Show: 'We couldn't find a face — please try a clearer photo with better lighting.' Never a technical error.

3.3 Step 3 — Check & Adjust

Let user verify AI result and make minor corrections. AI has already done the heavy work.

UI Element	Detail
Photo preview	Shows cropped result with a blue selection box around the face. User can drag the box or drag the photo to reposition.
Rotate button	Rotates 90° clockwise. Label: '■ Rotate'
Zoom in / out	Two buttons. Simple + and – zoom.
Brightness slider	Labeled 'Brightness'. Default: middle. No numbers shown.
Contrast slider	Labeled 'Contrast'. Default: middle.

UI Element	Detail
Sharpness slider	Labeled 'Sharpness'. Default: slightly above middle.
Reset button	Label: 'Reset to auto'. Returns all sliders to AI-chosen defaults.
Save & Next	Large primary button. Locks in crop and moves to Step 4.

3.4 Step 4 — Background

Optionally change the background to white or light blue for official use.

UI Element	Detail
Background cards	3 large visual cards: 'Original' (photo as-is), 'White' (AI removed, white fill), 'Light blue' (AI removed, blue fill). Each shows before/after preview.
AI model note	If model not downloaded yet, show: 'Needs one-time 3-second setup'. No mention of MB or model names.
Custom color	Small 'Other color' link below the cards. Opens color picker.
Name & Date stamp	Checkbox: 'Print name and date on photo'. Shows text inputs for name + date if checked.

3.5 Step 5 — Print Sheet (Key Feature)

This is the main reason studios will choose this app. Multiple people, different counts, one sheet of paper.

UI Element	Detail
People list (left panel)	Each person = one row. Shows: thumbnail, name ('Person 1'), and a +/- counter with 'photos' label.
Add another person	Dashed button: '+ Add another person'. Opens inline upload for next person. Each person has their own photo, crop, and background.
Remove person	Small × button on each row. Confirms before removing.
Drag to reorder rows	Drag handle (■) on left of each row. Reorders fill order in sheet preview instantly.
Sheet size tabs	Small tabs: A4 / 4×6 inch / 5×7 inch. Defaults to A4.
Sheet preview (right)	Live grid showing all photos tiled. Photos fill left-to-right, top-to-bottom. Each photo shows person initial badge.
Drag photos in preview	User can drag any photo cell to swap positions. Enables custom arrangement on the final sheet.
Empty slots	If total count < sheet capacity, remaining slots show as grey empty squares. This is fine — not an error.
Count summary	Below sheet selector: '10 of 12 slots filled' in muted text. No DPI, no mm.
Generate button	Label: 'Generate →'. Renders final 300 DPI sheet.

3.6 Download Screen

Option	Label	File	Use Case
Left card	Single photo — for online applications	JPG	Passport Seva Portal, visa portals, digital submission
Right card	Print sheet — A4 with all photos	PDF	Take to printer or print at home
Print now	Button inside right card	—	Opens browser print dialog. No intermediate steps.
Make new photo	Below both cards	—	Clears session, returns to Step 1. Keeps template choice.

4. Plain Language Guide

Every visible string in the app must follow this table. No technical term from the 'Never say' column may appear anywhere in the UI — not in buttons, labels, errors, or tooltips.

Concept	Say This	Never Say This
Photo dimensions	India / USA size, UK/Europe size	51×51 mm, 600×600 px, 2×2 inch
Output resolution	(nothing — user doesn't need to know)	300 DPI, 72 DPI, resolution, dots per inch
Face alignment	We found your face ✓	Face detection failed, FaceMesh error, no landmarks
AI model download	Getting ready... (progress bar)	Downloading 180 MB model, ONNX runtime, RMBG-1.4
Background removal	AI removes background	Segmentation, alpha mask, matting
Head position	(handled automatically — never shown)	Head ratio 73%, crown-to-chin, eye height
File size limit	Use a smaller photo (under 20 MB)	File exceeds 20971520 bytes
Offline mode	Works without internet ✓	Service Worker active, PWA cached, IndexedDB
Processing time	Almost done...	Inference running, tensor computation
Empty print slots	3 empty spaces left on this sheet	3 of 12 slots unfilled at 300 DPI

5. Multi-Person Print Sheet — Detailed Spec

This feature is the main reason studios will choose this app over competitors. It saves paper and time.

5.1 How It Works

1. Each person added to the sheet is processed independently through Steps 2–4 (upload → crop → background).
2. The sheet layout engine collects all persons' processed photos and their requested counts.
3. It tiles photos left-to-right, top-to-bottom, person by person in order.
4. User can drag individual photo cells in the sheet preview to any position — swaps on drop.
5. User can drag the person rows in the left panel to reorder whole groups.
6. Download renders the final arrangement at 300 DPI into a single PDF page.

5.2 Sheet Capacity Reference

Sheet	Size	India/USA (51×51 mm)	UK/Europe (35×45 mm)	Canada (50×70 mm)
A4	210×297 mm	12 photos (4×3)	18 photos (5×4)	8 photos (4×2)
4×6 inch	102×152 mm	4 photos (2×2)	4 photos (2×2)	2 photos (2×1)
5×7 inch	127×178 mm	6 photos (3×2)	8 photos (3×2)	4 photos (2×2)

Gutter between photos: 2 mm. Sheet margin: 4 mm on all sides. Fixed — not user configurable.

5.3 Drag & Drop Behaviour

- Dragging a photo cell in the preview swaps it with the target cell — no overwrite, always a swap
- Dragging a person row reorders fill order — sheet preview updates instantly
- Empty slots cannot be dragged — only filled photo cells can be dragged
- On touch devices: long-press to initiate drag

5.4 Adding a Second Person — User Flow (Target: under 20 seconds)

1. User clicks '+ Add another person' in the left panel.
2. A new upload zone slides in within the left panel (not a new page).
3. User uploads photo. AI runs face detection inline — small spinner shows.
4. After detection: background option shows as 3 small icon buttons (keep / white / blue). One tap.

5. Person is added to the list. Sheet preview updates automatically.
6. User sets count with + / – buttons.

6. Photo Size Templates

Stored as typed JSON config. Users never see any numbers — they only see the 'User Label' column.

User Label	Size (mm)	Print px @ 300 DPI	Background	Countries
India / USA (default)	51 × 51	602 × 602	White	India, USA, Philippines
UK / Europe	35 × 45	413 × 531	Light grey / White	UK, Germany, France, Japan, Australia
Canada	50 × 70	591 × 827	White	Canada
China	33 × 48	390 × 567	White	China
UAE / Saudi	35 × 45	413 × 531	White	UAE, Saudi Arabia, Gulf countries
Custom	User enters cm	Calculated	User picks	Any

7. AI Features — What They Do & How

7.1 Face Detection & Auto-Crop (MediaPipe FaceMesh)

Runs automatically after every upload. User never manually triggers it.

What AI Does	How (for developers)	User Sees
Finds the face	MediaPipe FaceMesh: 468 landmarks. ~8 MB WASM model, cached after first load.	'Finding your face...' spinner
Identifies face edges	Chin: landmark #152. Crown: #10 extrapolated +15%. Eyes: avg of landmark clusters.	Face box appears on photo
Calculates crop box	$\text{head_height} / \text{target_ratio} = \text{canvas_height}$. Canvas centered on face, offset for correct eye position.	Photo auto-crops to passport size
Falls back gracefully	If FaceMesh fails: show manual crop mode. User draws box themselves.	'We couldn't find a face — crop it manually'

7.2 Background Removal (Transformers.js + RMBG-1.4)

Optional. Only runs if user selects White or Light Blue background.

What AI Does	How (for developers)	User Sees
Removes background	RMBG-1.4 via @huggingface/transformers. Runs in Web Worker. WebGPU if available, WASM fallback for iOS.	'Getting ready...' then background disappears
Fills chosen color	Alpha mask applied to canvas, solid fill painted behind subject.	Clean white or blue background
First-time download	Model is ~180 MB. Downloaded once, cached in Service Worker. Subsequent uses: instant.	Progress bar: 'Getting ready...' (one time only)
Stays offline	Model cached in browser. No internet needed after first use.	Nothing — works same as always

7.3 Image Enhancement (Auto-applied on Upload)

- Auto-brightness: histogram analysis, corrects underexposed photos
- Auto-contrast: stretches histogram to eliminate washed-out look
- Auto-sharpness: light unsharp mask, improves blurry or scanned photos
- All enhancement uses Canvas 2D filters — no external library, instant processing
- User can override any enhancement with the sliders in Step 3

8. Technical Architecture

Everything runs in the browser. No server. No upload. No database. No cost per user.

8.1 Tech Stack

Layer	Technology	Why Chosen
Framework	Next.js 14 (App Router)	Static export, TypeScript, PWA-ready. Host free on Cloudflare Pages.
UI	React + Tailwind CSS + shadcn/ui	Fast to build. Accessible components. No extra cost.
Face detection	MediaPipe FaceMesh (WASM)	468 landmarks, works offline, 8 MB model, 5–15 ms per frame
BG removal	Transformers.js + RMBG-1.4	Best quality in-browser BG removal. WebGPU accelerated. WASM fallback on iOS.
Canvas editing	Fabric.js 6.x	Handles drag, resize, swap on print sheet canvas
Crop UI	react-easy-crop	Best crop UX. Free. Maintained.
Image resize	Pica (WebGL / Lanczos3)	High quality resize without server. Critical for 300 DPI output.
PDF export	jsPDF	Client-side PDF with correct DPI metadata. Free.
Offline storage	idb-keyval (IndexedDB)	Saves last project, settings, person photos. Simple promise API.
PWA / Offline	next-pwa (Workbox)	Service Worker caching of app shell + AI models after first load.
State	Zustand	Minimal global state. No Redux boilerplate.

8.2 Data Flow (Technical)

1. User uploads image → decoded to ImageBitmap → stored in memory
2. MediaPipe FaceMesh runs in Web Worker → returns landmark array
3. Auto-crop engine computes crop box → Pica resizes to 300 DPI target pixels
4. If BG removal: RMBG-1.4 in separate Worker → alpha mask → canvas composite
5. Print sheet: offscreen canvas at 300 DPI sheet size → all person photos composited
6. jsPDF: canvas toDataURL → PDF embed → Blob download

8.3 Print Quality Formula (For Developers Only)

Users don't see DPI. But if the math is wrong, the printed photo will be the wrong size. Use this formula everywhere.

```
outputPixels = Math.round( (dimensionMM / 25.4) × 300 )
```

- India/USA 51×51 mm → 602×602 px
- UK/Europe 35×45 mm → 413×531 px
- A4 sheet → 2480×3508 px
- 4×6 inch sheet → 1205×1795 px

Never use 96 DPI (screen resolution) for any print output canvas.

9. Offline & PWA Strategy

Feature	Works Offline?	Notes
Entire app UI	YES	Cached on first load via Service Worker
Photo upload & crop	YES	100% local — no server involved
Face detection (MediaPipe)	YES	8 MB WASM model cached after first use
Background removal (RMBG)	YES	Model cached after first use (one-time download)
PDF export	YES	jsPDF is bundled — no server needed
Print via browser	YES	window.print() is native
Settings & project save	YES	IndexedDB is local storage
Country template updates	NO	Fetches new templates silently in background when online

9.1 First-Load Experience

1. User visits the URL. App shell loads (~500 KB gzipped).
2. Service Worker installs in background. Caches app shell.
3. MediaPipe FaceMesh model (~8 MB): downloaded after first user interaction, not on page load. Cached permanently.
4. RMBG-1.4 model: downloaded ONLY if user picks White or Blue background. Not before. Shows 'Getting ready...' progress bar.
5. After these two one-time downloads: app is fully functional forever, with zero internet.

10. Development Milestones

#	Milestone	What Is Built	Days
M1	Project setup	Next.js 14 + TypeScript + Tailwind + shadcn/ui. next-pwa config. Deploy to Cloudflare Pages. GitHub Actions CI.	2
M2	5-step wizard shell	All 5 step screens with navigation, state management (Zustand). No AI yet — placeholder content.	2
M3	Upload + manual crop	Drag-drop upload, react-easy-crop, zoom/rotate, brightness/contrast/sharpness sliders. Works for one person.	3
M4	Size templates	JSON template config for 6 sizes. Step 1 card UI. mm→px conversion. Template persisted to localStorage.	2
M5	Layout engine + PDF	Sheet tiling algorithm. A4, 4×6, 5×7 support. jsPDF export at 300 DPI. Print button. ← MVP here	3
M6	AI face detection	MediaPipe FaceMesh WASM. Auto-crop on upload. Manual correction fallback. Web Worker integration.	4
M7	Multi-person sheet	Add/remove persons. Per-person count control. Drag rows to reorder. Drag photos in preview to swap.	4
M8	Background removal	Transformers.js + RMBG-1.4. Web Worker. WebGPU/WASM. Lazy download with progress. Color fill.	4
M9	Offline + storage	idb-keyval integration. Session save/restore. Settings persistence. PWA Service Worker fine-tuning.	2
M10	Polish + launch	All errors in plain language. Mobile responsive. Lighthouse ≥90. SEO. Sentry error monitoring.	3

Total: ~29 days solo. M1–M5 = working MVP (no AI) that can be launched on day 12.

11. Deployment (Zero Cost)

Item	Choice	Cost	Why
Hosting	Cloudflare Pages	Free	Unlimited bandwidth, global CDN, custom domain, automatic HTTPS
Domain	Already purchased	Paid	Point A/CNAME to Cloudflare Pages
CI/CD	GitHub Actions	Free	On push to main: build + deploy. Zero config for Next.js static export.
Error monitoring	Sentry free tier	Free	5,000 errors/month. Catches AI model failures, browser issues.
Analytics	Cloudflare Analytics	Free	Page views by country. No cookie consent needed. Privacy-friendly.
AI models	HuggingFace CDN (RMBG)	Free	Models served by HuggingFace. Cached locally after first use.

11.1 Deploy Steps

1. Push code to GitHub repository
2. Connect repository to Cloudflare Pages (takes 5 minutes)
3. Set build command: next build (with output: 'export' in next.config.js)
4. Add custom domain in Cloudflare Pages settings
5. Point domain DNS to Cloudflare Pages via CNAME record
6. HTTPS is automatic — no SSL certificate setup needed
7. Done. Every push to main auto-deploys in ~2 minutes.

12. What We Deliberately Left Out

Adding features is easy. Removing complexity is the hard part. Every item below was a conscious decision.

Feature	Removed ?	Reason
Head ratio percentage display	YES	Users don't know what this means. AI handles it silently.
DPI setting for users	YES	Always 300 DPI. No exceptions. No choice needed.
Compliance validator with warnings	YES	Overwhelming for non-tech users. AI auto-corrects silently.
Login / accounts	YES	Zero friction. No password. No email.
Cloud storage	YES	Privacy. Cost. Complexity. IndexedDB is enough.
Advanced photo editor (layers)	YES	Wrong product. This is a passport tool, not Photoshop.
Mobile-first native app	YES	PWA installs to home screen. No app store. No updates.
History of past projects	Later	Not needed for MVP. Studio operators make one photo at a time.
Batch upload 10 photos at once	Later	Multi-person sheet already solves batch printing. Phase 2.
Clothing change / virtual try-on	Later	Nice to have. Not core. Can be added after launch.

13. Error Messages — Plain Language Reference

Every error in the app uses the 'Show This Message' column. Developer error codes are logged to console only, never shown to users.

Situation	Show This Message	Show Technical Detail?
No face detected	We couldn't find a face in this photo. Try a clearer photo with the face directly facing the camera.	No
File too large	This photo is too large. Please use a photo under 20 MB.	No
Wrong file type	That file type doesn't work. Please use a JPG or PNG photo.	No
AI model loading slow	Getting ready... this only happens once.	No
AI model download failed	Couldn't set up the background feature. Check your internet and try again.	No — log to Sentry
PDF export failed	Something went wrong making the PDF. Try downloading as JPG instead.	No — log to Sentry
Storage full	Your browser storage is nearly full. Delete some old photos to continue.	No
Browser too old	Your browser doesn't support this app. Please update Chrome or try a different browser.	No

14. Quality Targets

What	Target	How to Verify
Full workflow time (upload → print sheet)	Under 60 seconds	Stopwatch test on mid-range Android with a clear portrait
Face detection success rate	≥ 90%	Test 50 photos: front-facing, good light, various skin tones
Photo dimensions accuracy	±1 px of 300 DPI target	Generate photo, measure pixels in image editor
Print size accuracy	±0.5 mm	Print on A4, measure with ruler
App load time (first load, 4G)	Under 3 seconds	Lighthouse / WebPageTest
App load time (cached, offline)	Under 1 second	DevTools throttling + Service Worker test
Lighthouse Performance	≥ 90	Chrome DevTools Lighthouse audit
Lighthouse Accessibility	≥ 95	Chrome DevTools Lighthouse audit
Browser support	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+	BrowserStack manual test

15. Risks & How We Handle Them

Risk	Impact	How We Handle It
RMBG model slow on old hardware	HIGH	Default is 'Original' — BG removal is opt-in. Show progress bar so user knows it's working.
WebGPU not available on iOS	MED	WASM fallback built into Transformers.js. Slower (10–30s) but functional.
Face detection fails (bad photo)	MED	Show clear message. Offer manual crop fallback. User draws box themselves.
Print scaling by OS changes photo size	HIGH	Add @media print CSS with explicit mm sizes. Test on Chrome, Firefox, Windows Print dialog.
Browser storage quota exceeded	MED	Compress saved blobs. Alert when >50 MB. Provide 'Clear saved photos' button.
jsPDF DPI not embedded correctly	HIGH	Unit test: generate PDF → parse metadata → assert 300 DPI. Catches before deploy.
Country standard changes	LOW	JSON config file. Takes 15 minutes to update and redeploy.

16. Pre-Launch Checklist

Functional

- Upload works: JPG, PNG, HEIC, WebP. Rejects wrong types with plain message.
- Face detection finds faces in 9 out of 10 test photos
- Manual crop fallback works when AI fails
- Background removal produces clean result on 5 test photos
- Print sheet: 2 people, different counts, drag to swap — works correctly
- PDF downloads and prints at correct size (measured with ruler)
- Single JPG downloads for online submission
- Settings remembered across browser restart

Offline

- Service Worker installs on first load
- Disconnect internet → reload → app works fully
- Face detection works offline (model cached)
- Background removal works offline after first use

Quality

- Lighthouse Performance ≥ 90
- Lighthouse Accessibility ≥ 95
- All error messages in plain language — zero technical jargon
- Tested on Chrome Android + Safari iOS
- Custom domain live with HTTPS
- Sentry error monitoring receiving test events
- No console errors on clean load

Content

- Privacy statement: 'Your photos never leave your device'
- No login, no cookies, no tracking
- SEO: title tag, meta description, Open Graph image
- robots.txt and sitemap.xml generated

The north star: a photo studio operator in Raipur who has never used Photoshop should be able to use this app without any help.